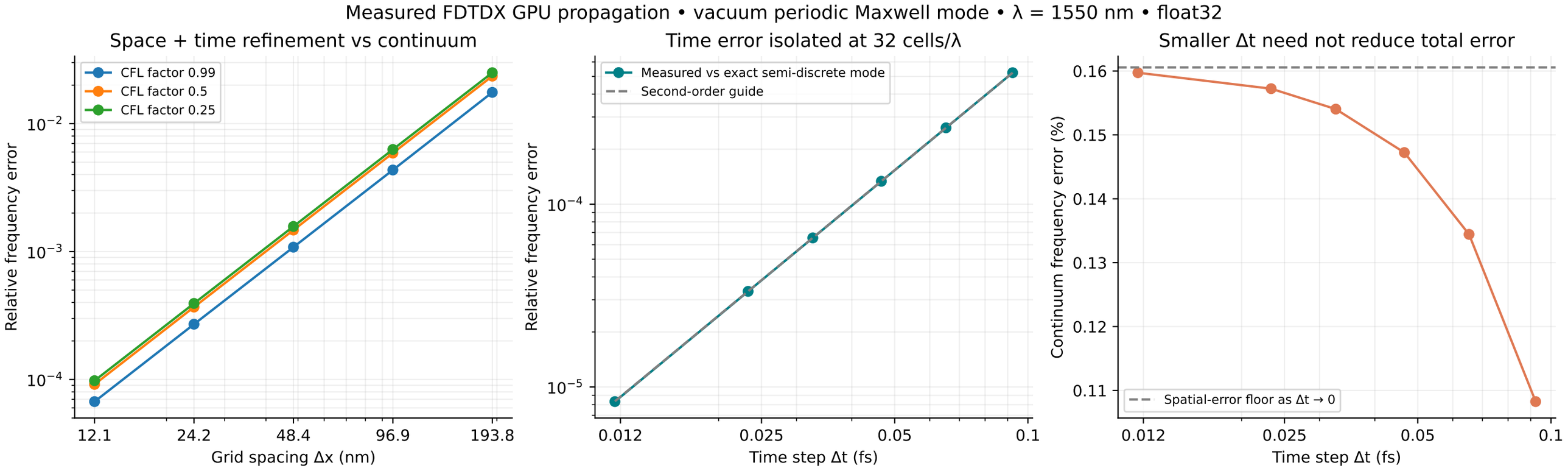
Measured FDTDX GPU propagation • vacuum periodic Maxwell mode • λ = 1550 nm • float32
Space + time refinement vs continuum
CFL factor 0.99
CFL factor 0.5
CFL factor 0.25
Relative frequency error
10^{-2}
10^{-3}
10^{-4}
12.1
24.2
48.4
96.9
193.8
Grid spacing Δx (nm)
Time error isolated at 32 cells/λ
Measured vs exact semi-discrete mode
Second-order guide
Relative frequency error
10^{-4}
10^{-5}
0.012
0.025
0.05
0.1
Time step Δt (fs)
Smaller Δt need not reduce total error
Continuum frequency error (%)
0.16
0.15
0.14
0.13
0.12
0.11
Spatial-error floor as Δt → 0
0.012
0.025
0.05
0.1
Time step Δt (fs)